Research Log

August 25, 2026 → Choosing the dataset

- **What I did:** Explored cBioPortal and searched the Datasets page for lung adenocarcinoma. Found three TCGA LUAD versions: PanCancer Atlas, Firehose Legacy and GDC.
- **What I learned:** The versions differ in how the mutations were called and how the clinical data were cleaned. Mixing them could give inconsistent results. The PanCancer Atlas version (`luad_tcga_pan_can_atlas_2018`) has harmonized survival data made specifically for outcome studies.
- **Problem:** I wasn't sure which version to use.
- **How I solved it:** Followed my mentor's advice to fix one dataset for the whole project: PanCancer Atlas.
- **Next:** Download the study ZIP.

August 27, 2026 → Downloading and opening the data

- **What I did:** Downloaded and unzipped the study, then found `data_mutations.txt` and `data_clinical_patient.txt`. Did not open the mutation file in Excel. Read both files with pandas using sep="\t", comment="#".
- **What I learned:** cBioPortal files start with metadata lines beginning with #, which have to be skipped. The mutation file has over 100 columns. The ones I need are `Hugo_Symbol`, `Tumor_Sample_Barcode`, `Variant_Classification`, `Variant_Type` and `HGVSp_Short`.
- **Problem:** With over 100 columns, it was hard to tell which ones mattered.
- **How I solved it:** Printed mutations.columns and mutations.head(), then checked them against the guide. I wrote my loading code with pathlib.Path, so the file paths are built in one place.
- **Next:** Extract the TP53 mutations.

September 1, 2026 → Extracting TP53

- **What I did:** Filtered the mutations to Hugo_Symbol == "TP53". Counted 318 records from 295 unique tumour samples. Printed the Variant_Classification breakdown: 195 missense, 55 nonsense, 24 + 3 frameshift, 23 splice site, plus a few others.
- **What I learned:** Mutation records are not the same as patients. 318 records but only 295 samples means some tumours have more than one TP53 mutation.
- **Problem:** Several categories in the breakdown (Splice_Region, Silent, In_Frame_Del) weren't in my guide's list.
- **How I solved it:** Mapped everything not in my list to "Other" for now, and noted that I should come back to it.
- **Next:** Look at the clinical file and work out how to link samples to patients.

September 3, 2026 → Clinical data and sample IDs vs patient IDs

- **What I did:** Explored the clinical file and found PATIENT_ID, OS_MONTHS, OS_STATUS, AGE and SEX. OS_STATUS is coded as 0:LIVING / 1:DECEASED.
- **What I learned:** The mutation file uses sample barcodes like TCGA-05-4382-01, but the clinical file uses patient IDs like TCGA-05-4382. The patient ID is the first three parts of the barcode. The "-01" means primary tumour.
- **Problem:** The two files can't be joined directly.

- **How I solved it:** Wrote sample_barcode_to_patient_id(), which splits the barcode on "-" and keeps the first three parts. All 295 samples mapped to 295 different patients.
- **Next:** Merge the TP53 table with the clinical table.

September 6, 2026 → Building the Phase 1 table

- **What I did:** Collapsed the TP53 records to one row per sample and added patient IDs. Left-joined onto the clinical table and filled patients without a TP53 record as "Wildtype". Wrote a QC checklist function that prints YES/blank for each item in the guide's Phase 1 checklist. Saved `phase1_final_table.csv`.
- **What I learned:** 566 patients in total: 295 mutant and 271 wildtype. 61 patients have no survival data.
- **Problem:** I wasn't sure that "no TP53 record" really means wildtype. It could also mean the tumour was never sequenced.
- **How I solved it:** Wrote this down as a question to check later (see Oct 5).
- **Next:** Count patients in each mutation type before doing any survival analysis.

September 10, 2026 → Group sizes and the grouping decision

- **What I did:** Counted patients per mutation-type combination. Missense 176, nonsense 51, frameshift 23, splice 19. Seven combinations had fewer than 10 patients, and my QC function flagged them.
- **What I learned:** Frameshift and splice groups are too small for reliable Kaplan–Meier curves. Each death would change the curve by about 5%.
- **Problem:** I had to decide how to merge the groups without being influenced by the survival results.
- **How I solved it:** Decided the groups before looking at survival: Wildtype/Missense only/Truncating-Splice only (nonsense + frameshift + splice)/Other-mixed. Wrote down my prediction first: truncating would be worst.
- **Next:** Learn Kaplan–Meier and the log-rank test.

September 15–20, 2026 → Learning survival analysis; first results

- **What I did:** Studied censoring, Kaplan–Meier curves, the log-rank test and hazard ratios. Ran my first survival analysis (`tp53_survival_analysis.py`) and made the first Kaplan–Meier figure.
- **What I learned:** Wildtype vs mutant: median OS 53.7 vs 42.3 months, log-rank p = 0.033. Three groups: p = 0.177, which is not significant. The truncating curve was lowest early on, but the missense and truncating curves overlapped a lot.
- **Problem:** My prediction was only partly supported. I wondered whether I'd done something wrong.
- **How I solved it:** Re-read the brief. An unexpected result is still a result, and I should ask why. I listed the brief's questions: sample size, coarse classes, p53 biology, stage, co-mutations.
- **Next:** Fit a Cox model.

September 27, 2026 → First Cox models

- **What I did:** Fitted Cox models for mutant vs wildtype, adjusted for age and sex.

- **What I learned:** Mutant vs wildtype HR = 1.37 (95% CI 1.03–1.84). Adjusting for age and sex barely changed it. A hazard ratio describes relative risk at any moment in time, which is different from the median survival.
- **Problem:** The brief says the key analysis is Survival ~ TP53 type + Stage. My Phase 1 table didn't keep the stage column.
- **How I solved it:** Noted that stage has to be added.
- **Next:** Add stage.

October 5, 2026 → Adding stage and verifying the data

- **What I did:**
  - Added AJCC pathologic stage. It comes from the TCGA Pan-Cancer Clinical Data Resource (Liu et al., Cell 2018), which is the source of cBioPortal's PanCancer Atlas clinical data.
  - Checked that the stage source matched my file patient by patient.
  - Checked the open question from Sept 6 using the study's cases_sequenced.txt case list.
- **What I learned:**
  - Survival times were identical once days were converted to months, and death status and sex matched for all 505 patients.
  - All 566 samples were sequenced, so "no TP53 record = wildtype" is valid.
  - Stage I was 54.5% of patients, because TCGA mostly collected tumours removed by surgery.
- **Problem:** Stage had sub-stages (IA, IB, IIIA…) and some entries said: "[Discrepancy]".
- **How I solved it:** Grouped stages into I–IV and treated "[Discrepancy]" or missing as unknown. 8 patients were affected.
- **Next:** Stage-adjusted Cox model.

October 8, 2026 → Investigating the "Other" category

- **What I did:** Went back to the "Other" variants I had set aside on Sept 1 and looked at their DNA changes (HGVSc).
- **What I learned:**
  - Silent variants: 4 records. They don't change the protein, but my code counted them as "Other". All 4 were in patients with another TP53 mutation, so no one's mutant/wildtype status was wrong. However, 2 missense and 2 nonsense patients were wrongly moved into "mixed" and dropped from the type comparison.
  - Splice_Region variants: 8 patients were "mutant" only because of these. Seven had the identical change at c.375, the last base of exon 4; the eighth was at c.993, the last base of exon 9. ClinVar lists c.375G>A as pathogenic because it breaks the splice site. The same spot being hit 7 times in 566 tumours suggests it matters.
- **Problem:** Should I change my classification after seeing the data?
- **How I solved it:** Fixed the Silent bug in the main analysis, because it was a coding error. I kept the splice-region decision as it was and tested the alternative only as a sensitivity analysis. Changing rules after seeing results is how researchers fool themselves.
- **Next:** Run the stage-adjusted models.

October 11, 2026 → Stage-adjusted results (the real "unexpected result")

- **What I did:** Fitted Survival ~ TP53 group + Stage, using the same patients for the unadjusted and adjusted models. Ran the Schoenfeld proportional-hazards test, a stratified model and three sensitivity analyses.
- **What I learned:**
    - Stage is the strongest predictor by far: HR 2.5–3.9 for stage II–IV.
    - Mutant vs wildtype fell from HR 1.37 to 1.21 (p = 0.22).
    - Missense fell to HR 0.99 because missense patients more often had stage II–III tumours. Truncating/splice stayed at HR 1.52 (p = 0.043).
    - Stage III failed the proportional-hazards test. The stratified model gave the same TP53 result.
    - Reclassifying splice-region variants moved the three-group p from 0.170 to 0.050. A handful of patients can change a borderline result.
- **Problem:** Is p = 0.043 "significant"? It's one of many tests, and the CI nearly touches 1.
- **How I solved it:** Calculated statistical power with Schoenfeld's formula. Only ~48% power for HR 1.5, needing ~250 deaths against my 117. Conclusion: consistent with the hypothesis, but not proof.
- **Next:** Clean up the code and write the report.

October 14, 2026 → Code clean-up and GitHub

- **What I did:**
    - Reorganised the project into three scripts (01_, 02_, 03_) with relative paths.
    - Removed duplicate function definitions I had accidentally pasted twice in `Analysis.py`.
    - Saved every number to `results/survival_results.json`.
    - Kept my first versions in `archive/`.
- **What I learned:** Reproducible code means someone else can download the data, run three commands and get my exact numbers.
- **Next:** Write the report.

October 18–20, 2026 → Writing the report

- **What I did:** Wrote the report. Followed the structure: question → why TP53 → why mutation type → data → classification → KM → log-rank → stage-adjusted Cox → interpretation → limitations → future directions.
- **What I learned:** The most interesting part of the project wasn't the p-values. It was realising that "TP53-mutant" isn't one category, and that stage, sample size and classification choices all shape the answer.
- **Next:** Validate in an independent cohort (e.g. MSK lung adenocarcinoma in cBioPortal) and test co-mutations (KRAS, EGFR, STK11, KEAP1).